Imperial maths

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Lecture 1 – 09/12/25

§1 Catenaries

Consider a chain of length L and constant density per unit length ρ_0, hanging from fixtures A and B with $AB < L$. What will the shape of the chain be as it hangs?

Let us make the centre of the chain, P, the origin. Then $B = (x_0, y_0)$.

Pick an arbitrary point on the chain, $Q = (x, y)$, and consider the piece of chain PQ. Let us consider the forces acting:

- We have a tension force $\mathbf{T}(x)$, tangent to the chain at Q.
- We have a constant tension T_0 tangent to the chain at P.
- We have the weight due to gravity of PQ.

Horizontally, we have

$$T_0 = T(x)\cos\theta \tag{1}$$

and vertically we have

$$T(x)\sin\theta = PQ\rho_0 g \tag{2}$$

where PQ is the length of the chain between P and Q. Assuming the shape is given by $y(x)$, we can calculate

$$PQ = \int_0^x \left(1 + y'^2\right)^{1/2} \, dx \tag{3}$$

Combining these equations, we have

$$\rho_0 g \int_0^x \left(1 + y'^2\right)^{1/2} \, dt = T_0 \tan\theta = T_0 y' \tag{4}$$

since $\tan\theta = \frac{dy}{dx}$. To eliminate the integral, let us differentiate both sides to obtain

$$y'' = \frac{\rho_0 g}{T_0}\left(1 + y'^2\right)^{1/2}$$

which is unfortunately non-linear. However, we might first wish to put $y' = p$, giving

$$\frac{dp}{dx} = \frac{\rho_0 g}{T_0}\left(1 + p^2\right)^{1/2}$$

and this is now a separable equation! We have

$$\int \frac{1}{(1+p^2)^{1/2}}\,dp = \int \frac{\rho_0 g}{T_0}\,dx$$

The right-hand side integral is simple, but for the left side, we need to use that

$$\cosh^2 t - \sinh^2 t = 1 \iff \cosh^2 t = 1 + \sinh^2 t$$

so we should substitute $p = \sinh t$, $dp = \cosh t\,dt$. We get

$$\begin{aligned}\int \frac{\cosh t}{(1+\sinh^2 t)^{1/2}}\,dt &= \int 1\,dt \\ &= t = \operatorname{arsinh} p \\ &= \operatorname{arsinh}\left(\frac{dy}{dx}\right)\end{aligned}$$

So we have

$$\operatorname{arsinh}\left(\frac{dy}{dx}\right) = \frac{\rho_0 g}{T_0}x + C$$

for some constant C. In fact, since $y' = 0$ at $x = 0$, $C = 0$. We have

$$\begin{aligned}\frac{dy}{dx} &= \sinh\left(\frac{\rho_0 g}{T_0}x\right) \\ \implies y &= \frac{T_0}{\rho_0 g}\cosh\left(\frac{\rho_0 g}{T_0}x\right) + D\end{aligned}$$

Since $x = 0$, $y = 0$, we get

$$D = -\frac{T_0}{\rho_0 g}$$

and hence we conclude

$$\boxed{y = \frac{T_0}{\rho_0 g}\left(\cosh\left(\frac{\rho_0 g}{T_0}x\right) - 1\right)}$$

In practice, we probably wish to do this in terms of the length. We have

$$\begin{aligned}\frac{L}{2} &= \int_0^{x_0}\left(1 + y'^2\right)^{1/2}\,dx \\ &= \frac{T_0}{\rho_0 g}y' && \text{(by 4)} \\ &= \frac{T_0}{\rho_0 g}\sinh\left(\frac{\rho_0 g}{T_0}x\right)\end{aligned}$$

so we can numerically solve for T_0 in terms of L, ρ_0 and g.

§2 Power series solutions of differential equations

Consider a simple differential equation:

$$\frac{dy}{dx} = y$$

with $y(0) = y_0$. We know the solution is given by $y = y_0 e^x$. But let's try solving this another, more generalisable way. Suppose

$$\begin{aligned} y &= \sum_{n=0}^{\infty} a_n x^n \\ \implies y' &= \sum_{n=1}^{\infty} n a_n x^{n-1} \end{aligned}$$

Thus we have

$$\begin{aligned} \sum_{n=0}^{\infty} a_n x^n &= \sum_{n=1}^{\infty} n a_n x^{n-1} \\ &= \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n \end{aligned}$$

Subtracting, we have

$$\begin{aligned} \implies \sum_{n=0}^{\infty} [a_n - (n+1)a_{n+1}] x^n = 0 \\ \implies a_n = (n+1)a_{n+1} \end{aligned}$$

for all n. From this it becomes clear that

$$a_n = \frac{a_0}{n!}$$

so

$$\begin{aligned} y &= \sum_{n=0}^{\infty} \frac{a_0}{n!} x^n \\ &= a_0 \sum_{n=0}^{\infty} \frac{1}{n!} x^n \\ &= a_0 e^x \end{aligned}$$

and the initial condition gives $a_0 = y_0$ as required.